

Introduction to the Internet of Things (IO3031)

Date: 9th April, 2026

Course Instructor

Ms. Abeer Bashir

Sessional-II Exam

Total Time (Hrs): 1

Total Marks: 30

Total Questions: 3

Roll No

Section

Student Signature

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Instructions:

1. Do not forget to write your Name and Roll Numbers.
2. Make assumptions if you think some data is missing.

CLO # 2: Apply the knowledge of Internet Connectivity Principles, IoT Networking Technologies, Protocols, and standards (PLO 1,C3)

Q1:

- a. An IoT system is developed using an ESP32 or ESP8266, where the device connects to a cloud platform (e.g., Blynk, Firebase) over Wi-Fi, as performed in the lab. The device periodically transmits the string "Internet of Things" to the server, where it is stored. **Determine and list the protocols used at each layer of the communication system between the IoT device and the cloud server.** [5+5 marks]

Layer	IoT Device	Server
Application Layer		
Transport Layer		
Network Layer		
Link Layer		

HTTP HTML
TCP UDP

- b. Examine the differences between traditional networks and software-defined networks (SDN). Illustrate your answer with appropriate labeled diagrams and list three key differences.

Centralized control plane
Flexible network config.
Auto config.

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Q2: An IoT-based smart parking system is deployed in a large parking area, where the area is divided into multiple slots. Each slot is equipped with sensors to monitor its status (empty or occupied). The sensors periodically detect the status and transmit this information reliably to a cloud server for monitoring. The cloud service provider offers three communication protocols: HTTP, MQTT, and CoAP, on a paid basis. The cost of using these protocols is as follows: [10 marks]

- HTTP (highest cost)
- MQTT (moderate cost)
- CoAP (lowest cost)

mqtt TCP low power
pub sub
reliable

Considering the large parking area with many slots, an efficient system is required that ensures relatively low cost, minimal energy consumption, and reliable end-to-end data communication. **Determine and select** the most suitable protocol for this system. Justify your selection with three valid reasons.

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Q3: For the given Low-power and Lossy network (LLN), construct DODAG. The Objective Function is as follows: Metric: Link cost, Rank=Hop count. **Construct** the complete DODAG, providing detailed calculations and explanations for each step until the final structure is established. [10 marks]

